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INTRODUCTION TO COMPUTER NETWORKS

Weighted Sum Rate Maximisation in RSMA

TEAM - 3 :

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AGENDA

- **Focus:** We're exploring Weighted Sum Rate Maximization (WSR) in Rate-Splitting Multiple Access (RSMA).
- **RSMA:** Splits messages into common and private streams improves efficiency, privacy, and reduces interference.
- **Objective:** Maximize data rates across users while meeting power and privacy constraints.
- **Techniques:** We'll discuss algorithms like ADMM, Gradient Descent, and Adam Optimizer to achieve these goals.
- **Takeaway:** RSMA with WSR is key to building faster, fairer, and more secure communication networks.

INTRODUCTION

- Rate-splitting multiple access is an advanced multiple access technique that improves spectral efficiency and user fairness in communication systems.
- In RSMA, each user's message is split into common and private streams:
    1. The common stream is decoded by all users.
    2. The private stream is decoded only by the intended user, ensuring security and tailored communication.
- RSMA effectively mitigates multi-user interference (MUI) and improves resource allocation compared to traditional methods like OFDM or TDMA.

OBJECTIVE

- **Maximize the Weighted Sum Rate (WSR) for all users, subject to the secrecy rate constraints of each user.**
- **Given: A user weight vector** $u = [u_1, \ldots, u_k]$
- **Optimization Problem:** $max\,(c,p) \sum_{k \in K} u_k \left(C_k + R_{p,k}\right)$
- **Subject to:**
  - **Secrecy Rate Constraint:** $R_{s,k} \geqslant R_{s,k}^{th}, k \in K$, **where** $R_{s,k}^{th}$ **is the secrecy rate threshold for each private stream.**
  - **Common Stream Decoding:** $\sum_{j \in K} C_j \leqslant R_{c,k}, k \in K$. **Ensures each user can successfully decode the common stream.**
  - **Transmit Power Constraint:** $tr\left(PP^H\right) \leqslant P_t$ **where** P_t **is the maximum transmit power.**
  - **Non-Negative Rates:** $c \geqslant 0$, **where** $c = [C_1, C_2, \ldots, C_k]$ **is the common rate allocation vector specifying the rate of the common stream allocated to each user.**

Litreature Review

Ref. No.	Year of Publication, Journal Name	Technology used	Inference
[1]	2019, IEEE	Rate-Splitting Multiple Access (RSMA) with Coordinated Multi-Point Joint Transmission (CoMP JT)	Dynamic Interference Management, Optimization via WMMSE (weighted minimum mean squared error)
[2]	2020, IEEE	RSMA's layered message splitting approach	Improved Latency and Reliability, Enhanced Spectral Efficiency and Energy Efficiency
[3]	2021, IEEE	Linear successive allocation method	IWMMSE approach can be used in the MIMO BC

Litreature Review

Ref. No.	Year of Publication, Journal Name	Technology used	Inference
[4]	2022, IEEE	Successive convex optimization	Numerical results demonstrate that the proposed secure beamforming scheme outperforms the conventional multiuser linear precoding (MULP) technique.
[5]	2023, IEEE	SCA algorithm	RS beamforming design outperforms baseline schemes by better balancing message confidentiality and sum-rate performance.
[6]	2024, IEE	KKT Conditions	Optimal beamforming structure in 1-layer RSMA enables efficient WSR performance and can be broadly applied across RSMA frameworks and extended to SDMA, NOMA, and massive MIMO system.

System Model

Initial Setup for Weighted Sum Rate Algorithm

- System parameters :
  - Number of users
  - Number of antennas used
- Noise and power :
  - N_0(noise power level which affects the signal to noise ratio)
  - P_t(Total transmit power constraint)
- Channel and power matrices :
  - h_k(matrix representing the channel gains between the BS antennas and each user)
  - P (matrix representing the power allocated to each use)
- Rate allocation and utility parameters :
  - c (vector consisting the rate allocations of each user)
  - u (user-weight vector)
  - γ_k (vector representing target signal-to-interference-plus-noise ratio (SINR) values for each user, which influences the rate calculation)

Algorithms Used

1.1) Alternating Direction Method of Multipliers (ADMM)

- ADMM is used to solve optimization problems that can be decomposed into subproblems, making it easier to solve them iteratively.
- In ADMM, we split the original problem into smaller problems and iteratively solve them while coordinating them with a penalty (Lagrange multiplier) term.
- The objective is to maximize weight sum rates for users, subject to power constraints and rate allocation constraints.
- ADMM splits the problem into:
  - Power optimization subproblem (solve for P)
  - Rate allocation subproblem (solve for c).
  - Lagrangian multiplier update (coordinate the power and rate decisions).

Algorithms Used

1.2) Alternating Direction Method of Multipliers (ADMM)

- ADMM Algorithm Steps
  - <u>Step 1: Power Optimization</u>, Solving the power allocation matrix P by iteratively adjusting P to maximize the secrecy rate Rs, considering interference and noise.
  - <u>Step 2: Rate Allocation</u>, Solving the rate allocation vector c by adjusting it to meet the constraints while maintaining the optimal allocation of rates for users.
  - <u>Step 3: Auxiliary Variable</u>, Introduce an auxiliary variable z to enforce consistency between the rate allocation c and the power allocation P.
  - <u>Step 4: Lagrange Multiplier</u>, Update the Lagrange multipliers λ to enforce constraints and guide the optimization process toward the optimal solution.
- After a certain number of iterations, the algorithm outputs the optimized power matrix P and rate allocation vector c.

Algorithms Used

2.1) Gradient Descent

- Gradient Descent is an optimization algorithm used to minimize the loss function by iteratively adjusting the model parameters
- It uses the gradient (partial derivatives) of the loss function to guide the direction of optimization.
- The algorithm converges to a local minimum by taking steps in the opposite direction of the gradient.
- The goal is to maximize weighted data rates for users, subject to power constraints and rate allocation constraints.
- Gradient Descent solves this by:
  - Power allocation subproblem (solve for the power matrix P).
  - Rate allocation subproblem (solve for the rate allocation vector c).
  - Gradient update steps (iteratively adjust P and c to minimize the loss function).

Algorithms Used

2.2) Gradient Descent

- Gradient Descent Algorithm Steps:
  - <u>Step 1: Secrecy Rate Calculation</u>,
    - Calculating the secrecy rate Rs for each user based on the current power allocation and rate allocation.
    - The secrecy rate depends on interference, signal power, and noise, and is given by a logarithmic function.
  - <u>Step 2: Compute Gradients</u>,
    - Computing the gradients of the secrecy rate with respect to the power matrix P and the rate allocation vector c.
    - The gradients indicate how much each parameter should change to increase the secrecy rate.
  - <u>Step 3: Update Parameters</u>,
    - Update P and c using the computed gradients and a learning rate α.

Algorithms Used

2.3) Gradient Descent

- The learning rate controls the size of each step taken in the gradient direction.

- <u>Step 4: Constraint Check</u>,
  - After each update, checking to done to ensure if the power and rate allocation satisfy the constraints, such as the power constraint (total power Pt) and the rate non-negativity (rate allocation c≥ 0).
  - Minimize this total loss function during the iterations.

- After a certain number of iterations, the algorithm outputs the optimized power matrix P and the rate allocation vector c, which maximize the secrecy rate while satisfying the problem's constraints (power and non-negativity).

Algorithms Used

3.1) Adaptive Moment Estimation (ADAM) optimizer

- Adam Optimizer is an adaptive optimization algorithm that computes adaptive learning rates for each parameter by using both the first and second moments of the gradients.
- It combines the advantages of both Momentum and RMSProp
- The goal is to optimize the power allocation matrix P and the rate allocation vector c to maximize secrecy rates, subject to constraints like power limits and non-negativity of rates.
- Adam updates the parameters by incorporating momentum and adaptive learning rates.
- ADAM Optimizer Steps:
  - <u>Step 1: Compute Gradient</u>, Computing the gradients of the objective function with respect to P and c.
  - <u>Step 2: Compute First and Second Moments (m and v)</u>,
    - Adam uses two moment estimates for each parameter:

Algorithms Used

3.2) Adaptive Moment Estimation (ADAM) optimizer

- mP and mc : The first moment (mean of gradients).
- vP and vc: The second moment (mean of squared gradients).

▪ These moments are updated at each iteration using exponential decay rates.

○ <u>Step 3: Bias Correction</u>,

▪ The moment estimates are biased towards zero, especially during the initial time steps.

▪ Adam corrects this bias by computing the bias-corrected estimates.

○ <u>Step 4: Parameter Update</u>,

▪ Parameters like α (learning rate), p, c are updated and ϵ is a small constant added to prevent division by zero.

○ <u>Step 5: Loss and Penalty Calculation</u>,

▪ Calculating the secrecy rate and the associated penalties (secrecy rate penalty and total loss).

▪ The algorithm attempts to minimize this total loss over the iterations.

Algorithms Used

4.1) Stochastic Gradient Descent (SGD)

- SGD is an optimization technique used to minimize an objective function iteratively by updating model parameters.
- Unlike Batch Gradient Descent (which computes the gradient over the entire dataset), SGD computes the gradient using only a random mini-batch of the data in each iteration, making it faster and more suitable for large datasets.
- SGD helps by updating these parameters iteratively using the gradients computed from a random subset of data (mini-batch).

Algorithms Used

4.2) Stochastic Gradient Descent (SGD)

- SGD Algorithm Steps
  - <u>Step 1: Select a Mini-Batch</u>,
    - In each iteration, a random subset (mini-batch) of data indices is selected.
    - This subset is used to compute the gradient for the update.
  - <u>Step 2: Compute Gradients</u>,
    - For each mini-batch, the gradient of the objective function with respect to the parameters P and c is computed.
  - <u>Step 3: Update Parameters</u>,
    - Once the gradients are computed, the model parameters P and c are updated where the learning rate (α) controls the step size of each update.
    - This helps the model learn efficiently without overshooting the optimal values.

Algorithms Used

4.3) Stochastic Gradient Descent (SGD)

- <u>Step 4: Calculate Loss and Penalty</u>,
  - After each update, the secrecy rate and penalties are calculated.
  - The secrecy penalty is computed by comparing the computed secrecy rates with the threshold .
  - If the secrecy rate is below the threshold, a penalty is added to the loss function R_s^{th}
- <u>Step 5: Constraint Check</u>,
  - After the update, the algorithm checks whether the constraints are satisfied:
    - Power constraint: The total power should be less than or equal to the power limit Pt.
    - Non-negativity of rate allocation: The rate vector c should have non-negative values.

Results

```
Final optimized precoder matrix (P):
[[ 0.27172003 -0.26200241  0.40323715]
 [ 0.29679541  0.13455878 -0.35357267]
 [ 0.23359277 -0.01866967 -0.44420091]
 [-0.41907789  0.26605506 -0.01670907]
 [-0.33089227 -0.03535755  0.30532385]]

Final optimized rate allocation vector (c):
[-1.03666053 -0.13686414 -0.7137461   0.16523181 -0.48812615]
```

a) Final optimized precoding matrix (P) and rate allocation vector using ADMM Algorithm

Results

```
Final optimized precoder matrix (P):
[[-0.06707803 -0.43625016 -0.19457098]
 [ 0.42910439  0.25404525 -0.07190455]
 [-0.32322658 -0.37299967  0.37239635]
 [ 0.43040987  0.35257896 -0.19540822]
 [-0.08341797 -0.21961874 -0.27101289]]

Final optimized rate allocation vector (c):
[-0.29484631 -0.00162829 -1.0371219  -0.56375574 -0.28377946]
```

b) Final optimized precoding matrix (P) and resource allocation vector using Gradient descent

Results

```
Final optimized precoder matrix (P):
[[0.40943242 0.08882001 0.17778449]
 [0.00821679 0.62606539 0.46704749]
 [0.22793361 0.90517031 0.7595831 ]
 [0.03013148 0.09107413 0.71856684]
 [0.87196566 0.44284081 0.68203384]]

Final optimized rate allocation vector (c):
[0.26074116 0.84067535 0.16877088 0.84833645 0.96295826]
```

c) Final optimized precoding matrix (P) and resource allocation vector using ADAM optimizer

Results

```
Final optimized precoder matrix (P):
[[ 0.27898475 -0.28977227  0.12855576]
 [ 0.06288544  0.00181315 -0.53468427]
 [-0.09214736  0.30721889  0.43678616]
 [ 0.06745483  0.32081245 -0.13863476]
 [-0.36535223  0.37701303 -0.04025228]]

Final optimized rate allocation vector (c):
[-0.11223929 -0.03887542  0.08906072  0.18082549 -0.06623722]
```

d) Final optimized precoding matrix (P) and resource allocation vector using Stochastic Gradient descent

Conclusion

- After implementing ADMM, Adam Optimizer, Gradient Descent, and Stochastic Gradient Descent, it was observed that the Adam Optimizer yielded the best results, outperforming the other algorithms in terms of optimization performance.
- Unlike ADMM, Gradient Descent, and Stochastic Gradient Descent, the Adam Optimizer successfully satisfied all the constraints, including secrecy rate, power, and non-negativity conditions.
- This demonstrated its effectiveness in converging to an optimal solution while ensuring that all constraints were met, leading to more reliable and efficient optimization for secure communication.

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